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Topic 8 — Fuels and Earth Science

Edexcel GCSE (9-1) Chemistry (1CH0) — Paper 2

Most of the energy used for transport, heating and electricity still comes from burning **fossil fuels** made from crude oil and natural gas. This topic covers how crude oil is separated and processed into useful fuels, what happens when those fuels burn (and the pollutants formed), and then zooms out to the **Earth's atmosphere**: how it formed, how it changed, and how human activity is now changing the climate.

Topic 8 is common to GCSE Chemistry and Combined Science — there is no chemistry-only content in this topic, and no Higher-tier-only content. There is no core practical in Topic 8, but you should know the laboratory cracking experiment and the test for oxygen. The naming, formulae and reactions of individual alkanes and alkenes are covered in Topic 9 (chemistry-only).

8.1 Hydrocarbons and Crude Oil

> **Hydrocarbons** are compounds that contain **carbon and hydrogen atoms only**.

Methane (CH_4), octane (C_8H_{18}) and ethene (C_2H_4) are hydrocarbons. Ethanol (C_2H_5OH) is *not* — it contains oxygen.

> **Crude oil** is a **complex mixture of hydrocarbons**, containing molecules in which carbon atoms are joined in **chains** or **rings**.

- Crude oil is an important source of **useful substances**: **fuels** (petrol, diesel, kerosene, etc.) and **feedstock for the petrochemical industry** (raw materials for making plastics, solvents, detergents, medicines, dyes and lubricants).

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- Crude oil is a **finite resource**: it was formed over millions of years from the remains of ancient marine organisms (mainly plankton) buried in mud, under heat and pressure, and is being used up far faster than it can form. It is **non-renewable**.
- Because it is a mixture, the hydrocarbons in it are not chemically bonded to each other, so they can be separated by a physical process — fractional distillation.

Exam Tip: You do not need to know the names, formulae or structures of specific ring molecules in crude oil — just that crude oil contains both chain and ring hydrocarbons.

Homologous series

A **homologous series** is a series of compounds which:

- have the **same general formula**,
- differ by **CH_2** in molecular formulae from neighbouring compounds,
- show a **gradual variation in physical properties**, as exemplified by their boiling points,
- have **similar chemical properties** (because they have the same functional group).

The **alkanes** are the main homologous series in crude oil, general formula **C_nH_{2n+2}**.

Alkane	Formula	Boiling point / °C	State at room temperature
methane	CH_4	−162	gas
ethane	C_2H_6	−89	gas
propane	C_3H_8	−42	gas
butane	C_4H_{10}	−1	gas
pentane	C_5H_{12}	36	liquid
hexane	C_6H_{14}	69	liquid
octane	C_8H_{18}	126	liquid

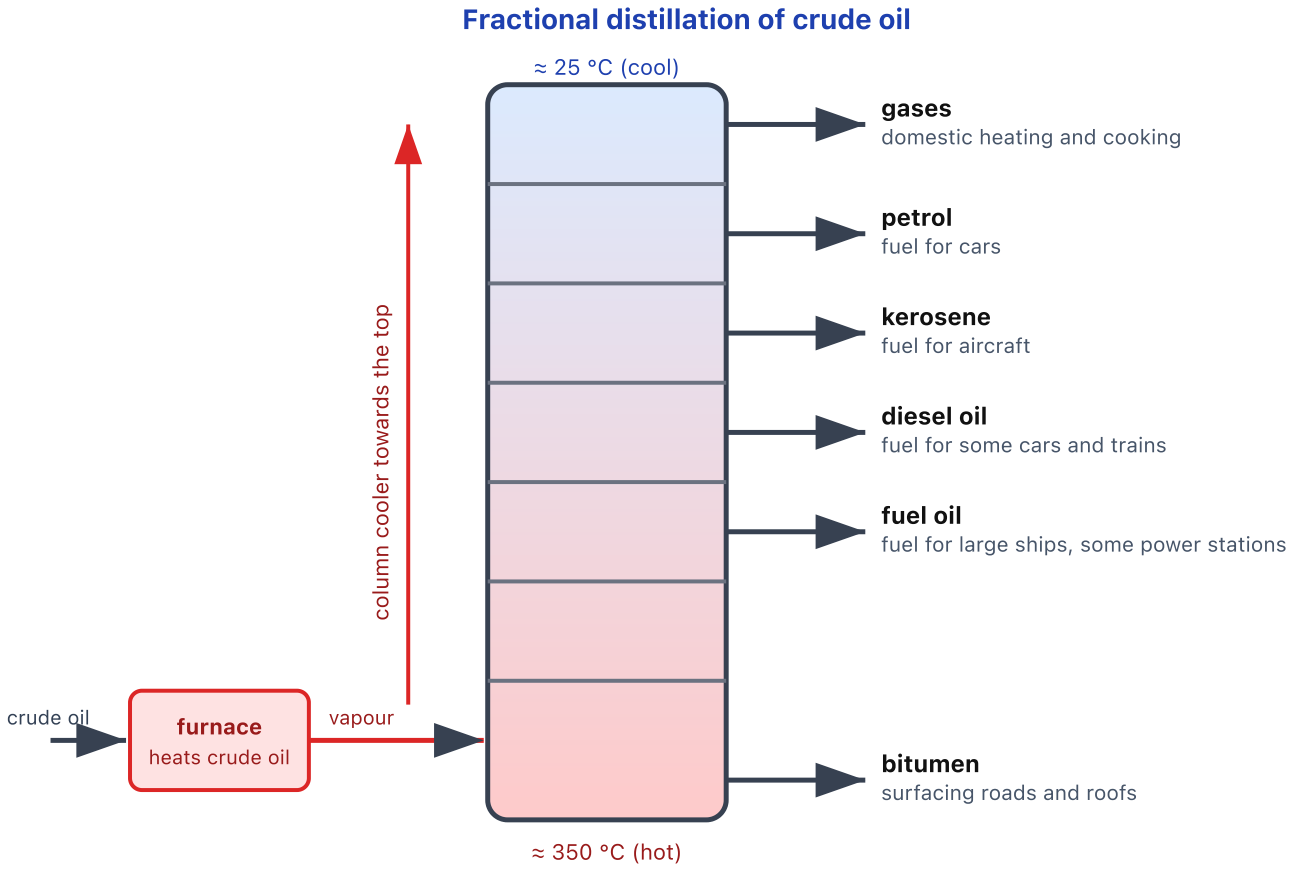
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Each compound differs from the next by CH_2, and the boiling point increases gradually as the chain gets longer.

> **Worked Example:** Use the general formula to find the formula of the alkane with 12 carbon atoms.
> C_nH_{2n+2} with n = 12: H = 2(12) + 2 = 26 → **$C_{12}H_{26}$**.

8.2 Fractional Distillation of Crude Oil

Crude oil is separated into simpler, more useful mixtures called **fractions** by **fractional distillation**. Each fraction is a mixture of hydrocarbons with a similar number of carbon atoms, so they have **similar boiling points**.



Lower down: bigger molecules, higher boiling point, more viscous, harder to ignite

How it works

1. Crude oil is **heated** in a furnace until most of it **evaporates** to a vapour.

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2. The vapour enters near the bottom of a tall **fractionating column** that is **hot at the bottom and cooler at the top** (there is a temperature gradient).
3. The vapours **rise** up the column and **cool**.
4. Each hydrocarbon **condenses** when it reaches the level where the temperature is just below its boiling point; the liquid is collected on trays and piped off as a fraction.
5. Hydrocarbons with **high boiling points** (large molecules) condense **low down**; hydrocarbons with **low boiling points** (small molecules) rise further and condense **near the top**. The smallest molecules stay as gases and leave at the top.
6. The very largest molecules don't evaporate at all and are collected at the bottom as **bitumen**.

> **Exam Tip:** Use the words **evaporate** (or vaporise) and **condense** correctly. The column separates by **boiling point**, which depends on **molecule size**. You do not need to know the names of specific fractionating columns.

Names and uses of the fractions

Fraction	Approx. carbon atoms	Use
Gases	1–4	domestic heating and cooking (bottled gas)
Petrol	5–10	fuel for cars
Kerosene	10–16	fuel for aircraft
Diesel oil	14–20	fuel for some cars and trains
Fuel oil	20–40	fuel for large ships and in some power stations
Bitumen	> 40	surfacing roads and roofs

Carbon-number ranges are approximate guides; you need the names, order and uses.

> **Exam Tip:** Memory aid for the order from top to bottom: **G**et **P**aid **K**eep **D**ancing **F**or **B**its — Gases, Petrol, Kerosene, Diesel, Fuel oil, Bitumen.

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8.3 How the Properties of Fractions Differ

Fractions differ in their properties because their molecules contain different numbers of carbon and hydrogen atoms. The **bigger** the molecules:

Property	Small molecules (e.g. gases, petrol)	Large molecules (e.g. fuel oil, bitumen)	Explanation
Boiling point	low	high	larger molecules have stronger **intermolecular forces** between them, so more energy is needed to separate them
Ease of ignition	easy to ignite (very flammable)	hard to ignite	smaller molecules are more volatile — they evaporate easily to form a flammable vapour
Viscosity (how thick / how hard to pour)	low — runny	high — thick, flows slowly	longer molecules have stronger intermolecular forces and can tangle, so they flow past each other less easily

Exam Tip: It is the forces **between** molecules (intermolecular forces) that are overcome when hydrocarbons boil — **not** the covalent bonds inside the molecules. Saying "covalent bonds break when it boils" is a classic lost mark.

8.4 Combustion of Hydrocarbon Fuels

Complete combustion

Complete combustion of a hydrocarbon fuel happens when there is a **plentiful supply of oxygen**. The products are **carbon dioxide and water**, and energy is given out (exothermic).

hydrocarbon + oxygen → carbon dioxide + water

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

$$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$$

The carbon and hydrogen in the fuel are **oxidised**. Complete combustion gives a clean blue flame and releases the maximum amount of energy from the fuel.

Worked Example: Balance the equation for the complete combustion of octane, C_8H_{18}.
$C_8H_{18} + O_2 \rightarrow 8CO_2 + 9H_2O$ (balance C then H). Oxygen atoms on the right = 16 + 9 = 25, so 12½ O_2.
Double everything to remove the half: **$2C_8H_{18} + 25O_2 \rightarrow 16CO_2 + 18H_2O$**.

Incomplete combustion

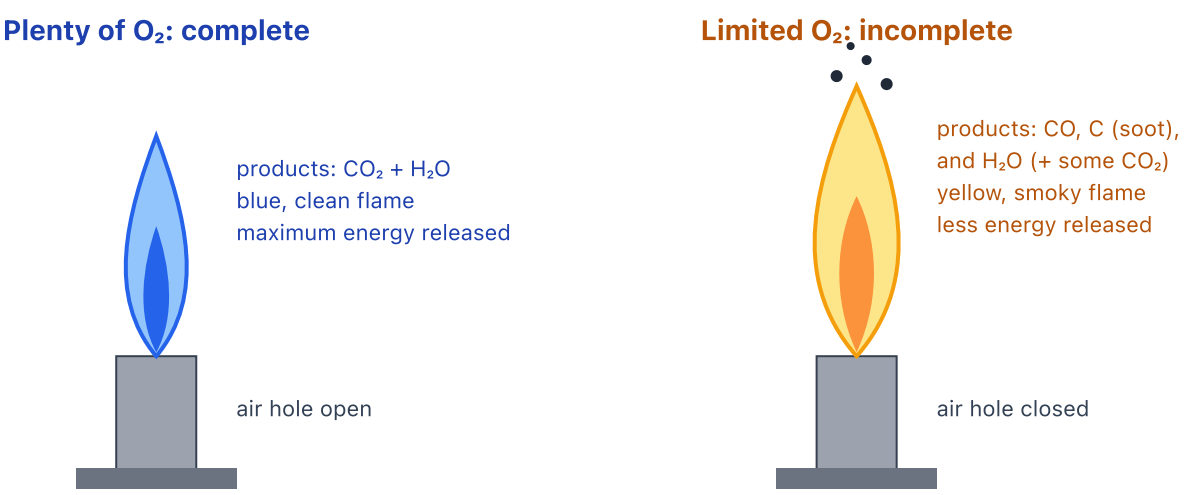
If the **supply of oxygen is limited**, there is not enough oxygen to oxidise all the carbon to CO_2. **Incomplete combustion** produces **carbon monoxide (CO)** and/or **carbon (soot)**, as well as water. Less energy is released.

$$2CH_4 + 3O_2 \rightarrow 2CO + 4H_2O \quad \text{(carbon monoxide formed)}$$

$$CH_4 + O_2 \rightarrow C + 2H_2O \quad \text{(carbon / soot formed)}$$

Incomplete combustion often gives a smoky, **yellow flame** (glowing soot particles).

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Why carbon monoxide is toxic

- Carbon monoxide is a **colourless, odourless** gas, so it cannot be detected by our senses.
- When breathed in, CO **combines with haemoglobin** in red blood cells (more strongly than oxygen does).
- This **reduces the capacity of the blood to carry oxygen** around the body, leading to drowsiness, headaches, unconsciousness and eventually death.

Problems caused by incomplete combustion in appliances

- Faulty or poorly ventilated **gas fires, boilers and cookers** can produce **carbon monoxide** — this is why homes should have **CO detectors** and appliances should be serviced regularly.
- **Soot** (carbon particles) blackens appliances and buildings, can **block flues and chimneys** (so waste gases, including CO, cannot escape) and causes breathing problems if inhaled.
- Less energy is released from the fuel, so incomplete combustion **wastes fuel** and money.

> **Exam Tip:** For "why does incomplete combustion produce CO?" the answer is: there is **insufficient oxygen** to oxidise all the carbon in the fuel fully to carbon dioxide. Don't confuse CO (carbon monoxide, toxic) with CO_2 (carbon dioxide, a greenhouse gas).

8.5 Other Pollutants from Fuels

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Sulfur dioxide and acid rain

Some hydrocarbon fuels (especially coal, diesel and fuel oil) contain **sulfur compounds as impurities**. When the fuel is burned, the sulfur is oxidised to **sulfur dioxide**:

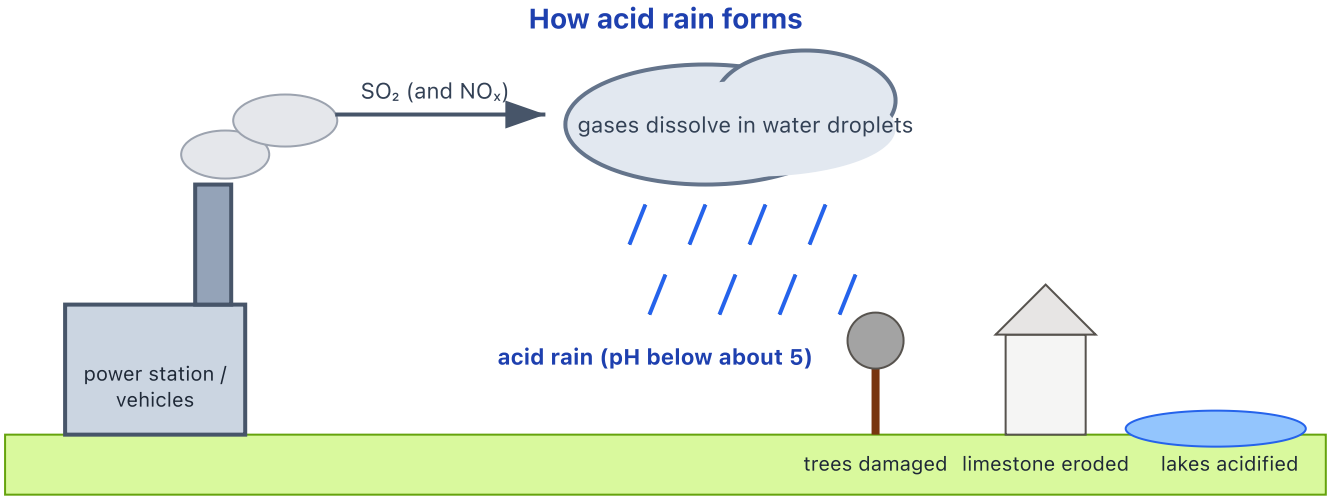
$$S + O_2 \rightarrow SO_2$$

Sulfur dioxide dissolves in rainwater to form an acidic solution — **acid rain**. (In the air it can be further oxidised, forming sulfuric acid, H_2SO_4.)

Problems caused by acid rain:

- **Damages buildings and statues** made of limestone or marble (calcium carbonate reacts with acid) and **corrodes metal** structures.
- **Acidifies lakes and rivers**, harming or killing fish and other aquatic life.
- **Damages trees and plants** (leaves are damaged and nutrients are leached from soil).
- SO_2 itself irritates the lungs and can trigger asthma.

Reducing SO_2: remove sulfur from fuels before they are sold (low-sulfur petrol and diesel), and "scrub" waste gases from power stations (e.g. with calcium oxide/carbonate).



Oxides of nitrogen

When fuels are burned in **engines**, the temperature is very high. At these high temperatures, **nitrogen and oxygen from the air** (not from the fuel!) combine to form **oxides of nitrogen** (NO_x), such as nitrogen monoxide (NO) and nitrogen dioxide (NO_2):

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$$N_2 + O_2 \rightarrow 2NO \qquad 2NO + O_2 \rightarrow 2NO_2$$

- Nitrogen is normally very unreactive (strong N≡N triple bond), which is why the reaction only happens at the high temperatures inside an engine.
- Oxides of nitrogen cause **acid rain**, **photochemical smog**, and breathing problems (they irritate the lungs and worsen asthma).
- **Catalytic converters** on car exhausts convert NO_x back to N_2 (and CO to CO_2).

> **Exam Tip:** The nitrogen in NO_x comes from the **air**, not from the fuel. Hydrocarbon fuels don't contain nitrogen. Say: "the high temperature in the engine provides enough energy for nitrogen and oxygen from the air to react".

Summary of pollutants

Pollutant	How it forms	Problem caused
Carbon dioxide, CO_2	complete combustion of any carbon-based fuel	greenhouse gas → climate change
Carbon monoxide, CO	incomplete combustion (limited O_2)	toxic — reduces blood's ability to carry oxygen
Carbon (soot)	incomplete combustion (very limited O_2)	blackens buildings, blocks flues, breathing problems; global dimming
Sulfur dioxide, SO_2	sulfur impurities in fuel are oxidised	acid rain; respiratory problems
Oxides of nitrogen, NO_x	N_2 and O_2 from air react at high engine temperatures	acid rain; smog; respiratory problems

8.6 Hydrogen as an Alternative Fuel

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Hydrogen can be used as a fuel for cars, either **burned in a combustion engine** or used in a **hydrogen–oxygen fuel cell** to generate electricity for an electric motor (fuel cells are covered in Topic 5).

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$$

You must be able to **evaluate the advantages and disadvantages of using hydrogen, rather than petrol, as a fuel in cars**.

Advantages of hydrogen	Disadvantages of hydrogen
The only product of combustion is **water** — no CO_2 from the car, so no contribution to climate change at the point of use.	Hydrogen is a **gas** — it takes up a large volume, so it must be stored under **high pressure** (or liquefied at very low temperature) in heavy, strong tanks.
No CO, soot or SO_2 produced (no carbon or sulfur in the fuel).	It is highly **flammable/explosive** — leaks are a safety risk.
Releases **more energy per kilogram** than petrol.	Few **filling stations** supply hydrogen; vehicles are expensive.
Can be made from water (a renewable source), e.g. by electrolysis.	Most hydrogen is currently made from **natural gas** (methane) or by electrolysis using electricity from fossil fuels — which produces CO_2 elsewhere.
Fuel cells are more efficient than combustion engines and quiet.	Burning hydrogen in an engine at high temperature can still produce some **NO_x** from air.

Exam Tip: "Evaluate" questions need **both sides** and then a **justified conclusion**. A strong conclusion links the two, e.g. "hydrogen is only a low-carbon fuel overall if the hydrogen is made by electrolysis using electricity from renewable sources".

8.7 Fossil Fuels Are Non-Renewable

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- **Petrol, kerosene and diesel oil** are **non-renewable fossil fuels** obtained from **crude oil**.
- **Methane** is a non-renewable fossil fuel found in **natural gas**.
- Coal is also a fossil fuel. Fossil fuels formed over millions of years and are being used up much faster than they are replaced — they will eventually run out.
- As supplies decrease, they become harder and more expensive to extract. Crude oil is also needed as a **feedstock** for plastics and other chemicals, which is another reason not to burn it all.

8.8 Cracking

> **Cracking** involves the **breaking down of larger, saturated hydrocarbon molecules (alkanes)** into **smaller, more useful ones**, some of which are **unsaturated (alkenes)**.

Cracking is a **thermal decomposition** reaction. It is carried out by passing the vapour of a large alkane over a hot **catalyst** (e.g. aluminium oxide or a zeolite) at a **high temperature** (several hundred °C).

$$C_{10}H_{22} \rightarrow C_8H_{18} + C_2H_4 \quad \text{(decane} \rightarrow \text{octane + ethene)}$$

$$C_{12}H_{26} \rightarrow C_8H_{18} + C_2H_4 + C_2H_4$$

$$C_{14}H_{30} \rightarrow C_6H_{14} + 4C_2H_4$$

Worked Example: Complete the cracking equation: $C_{16}H_{34} \rightarrow C_{10}H_{22} +$ _____
Carbon: 16 – 10 = 6; hydrogen: 34 – 22 = 12. The missing product is **C_6H_{12}** (an alkene, since it fits C_nH_{2n}). Alternatively it could be written as $3C_2H_4$.

Exam Tip: Cracking equations must balance: the total C and H atoms on each side are equal. At least one product must be an alkene (because there are not enough H atoms for all the products to be alkanes). Alkanes fit C_nH_{2n+2}, alkenes fit C_nH_{2n}.

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Why cracking is necessary

- Fractional distillation produces **more** of the large-molecule fractions (e.g. fuel oil, diesel) than there is demand for, and **not enough** of the small-molecule fractions such as **petrol**, which are in high demand.
- Cracking converts the surplus long-chain hydrocarbons into **shorter-chain hydrocarbons** that match demand — making the most of a finite resource.
- Cracking also produces **alkenes** (e.g. ethene, propene), which are reactive and are used as feedstock to make **polymers** (plastics) and other chemicals, such as ethanol.

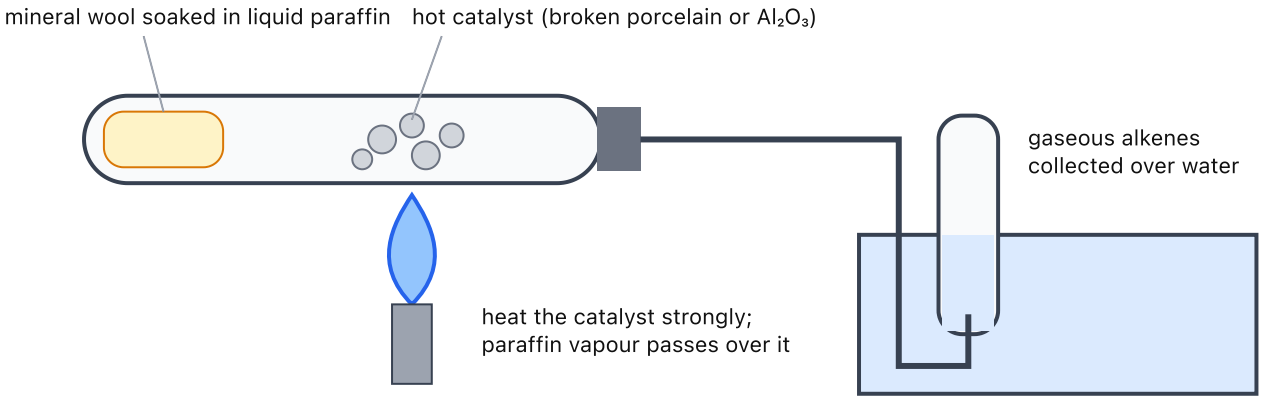
Fraction	Supply from crude oil (%)	Demand (%)
Gases	3	5
Petrol	12	25
Kerosene	14	9
Diesel oil	20	23
Fuel oil and bitumen	51	38

Illustrative data only. Exam questions often give a supply-and-demand table or bar chart: the fractions where supply > demand are the ones that are cracked; those where demand > supply are the ones cracking helps to produce.

Cracking in the laboratory

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Cracking liquid paraffin in the laboratory



- The catalyst is heated strongly, then the paraffin is warmed so its vapour passes over the hot catalyst and cracks.
- The gas collected (mainly ethene and other small alkenes) **decolourises bromine water** (orange → colourless), showing it is unsaturated.
- **Safety:** remove the delivery tube from the water **before** stopping heating, to prevent **suck-back** of cold water into the hot tube (which could crack it). Discard the first tube of gas (it is mainly air).

8.9 The Earth's Early Atmosphere

Scientists' ideas about the early atmosphere are based on **evidence** such as the gases released by volcanoes today, the composition of the atmospheres of other planets (e.g. Mars and Venus, mainly CO_2), the chemistry of ancient rocks and fossils, and air trapped in ice cores — but the evidence is limited because it was billions of years ago, so the ideas are a **theory** that may change.

- The **gases produced by volcanic activity formed the Earth's early atmosphere**.
- The early atmosphere is thought to have contained:
  - **little or no oxygen**,
  - a **large amount of carbon dioxide**,
  - **water vapour**,
  - **small amounts of other gases** (e.g. nitrogen, ammonia, methane).

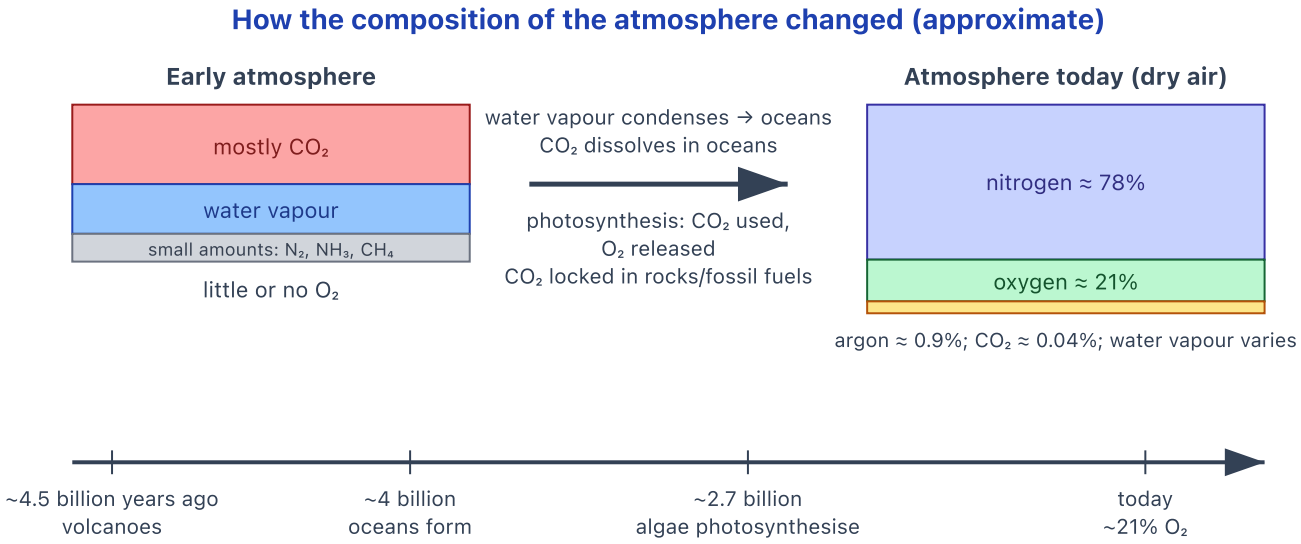
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- You must be able to **interpret evidence** relating to this — e.g. rocks containing iron(II) compounds (not iron(III) oxides) formed early in Earth's history suggest there was little oxygen; later "red beds" containing oxidised iron show oxygen had appeared.

How the atmosphere changed

1. **Formation of the oceans:** as the Earth cooled, the **water vapour condensed** to form liquid water, which collected to form the **oceans**.
2. **CO_2 dissolved in the oceans:** the amount of carbon dioxide in the atmosphere **decreased** as it **dissolved** in the oceans as they formed. Some of this dissolved carbon dioxide was later locked up in **carbonate** rocks (e.g. limestone, formed from shells of marine organisms) and in fossil fuels.
3. **Photosynthesis added oxygen:** the growth of **primitive plants** (algae, then plants) **used carbon dioxide and released oxygen** by photosynthesis. Consequently the amount of oxygen in the atmosphere **gradually increased** and CO_2 decreased further.

$$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2 \quad \text{(photosynthesis)}$$



Dates are approximate and uncertain — they are not required by the specification.

Composition of today's atmosphere

Dry air today is approximately **78% nitrogen**, **21% oxygen**, **0.9% argon** (and small amounts of other noble gases) and about **0.04% carbon dioxide**, plus a variable amount of water vapour.

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Test for oxygen

> **Test for oxygen:** insert a **glowing splint** into the gas — oxygen **relights the glowing splint**.

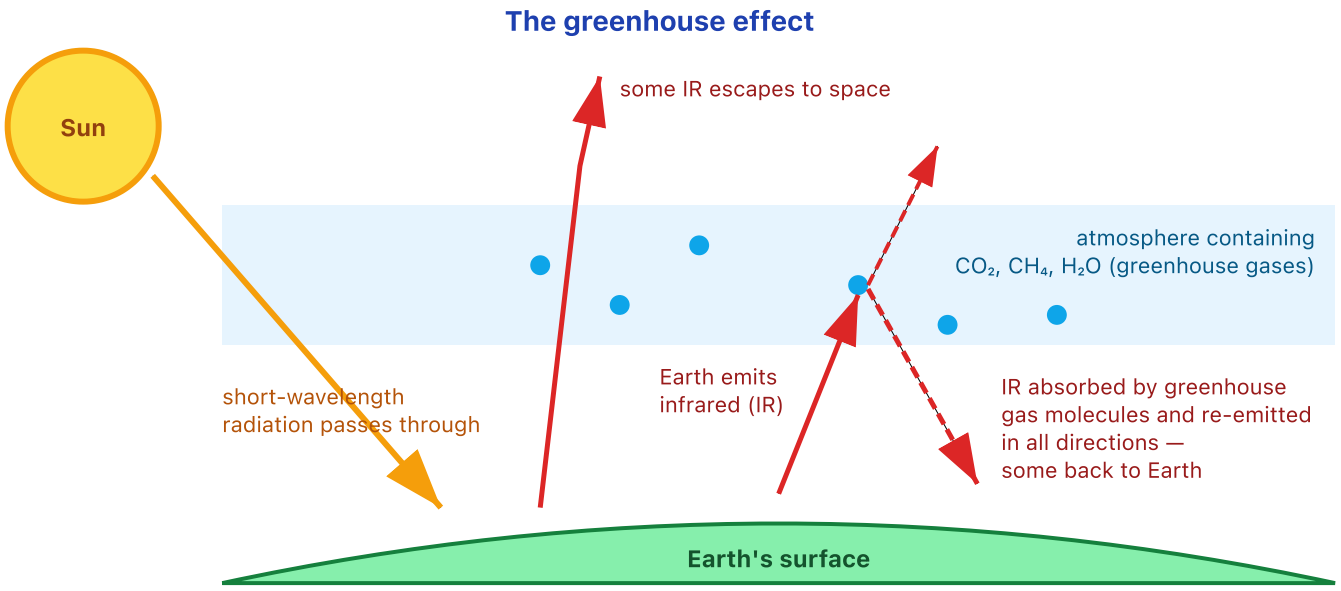
> **Exam Tip:** Don't mix up the gas tests: oxygen **relights a glowing** splint; hydrogen gives a **squeaky pop with a lighted** splint; carbon dioxide turns **limewater milky/cloudy**; chlorine **bleaches damp litmus paper**.

8.10 The Greenhouse Effect

The greenhouse effect keeps the Earth warm enough for life. Without any greenhouse gases the average surface temperature would be about –18 °C rather than about +15 °C.

1. The Sun emits radiation, mostly **short-wavelength** (visible light and ultraviolet), which passes through the atmosphere and is absorbed by the Earth's surface, warming it.
2. The warm Earth radiates energy back out as **longer-wavelength infrared radiation**.
3. **Greenhouse gases** in the atmosphere — **carbon dioxide, methane and water vapour** — **absorb** some of this infrared radiation and then **re-emit** it in all directions, including back towards the Earth.
4. This reduces the amount of heat lost to space, keeping the lower atmosphere and surface warmer.

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Exam Tip: Greenhouse gases **absorb and re-emit infrared** radiation — they do not "reflect" heat, and the greenhouse effect is **not** caused by the hole in the ozone layer. Name the three greenhouse gases in the specification: carbon dioxide, methane and water vapour.

8.11 Human Activity and Climate Change

Human activities increase the amounts of greenhouse gases in the atmosphere:

Greenhouse gas	Human sources
Carbon dioxide	burning fossil fuels (power stations, vehicles, heating); deforestation (fewer trees to absorb CO_2, and burning the wood releases CO_2); cement production
Methane	cattle and other livestock (digestion), rice paddy fields, landfill sites (decomposing waste), leaks from gas extraction

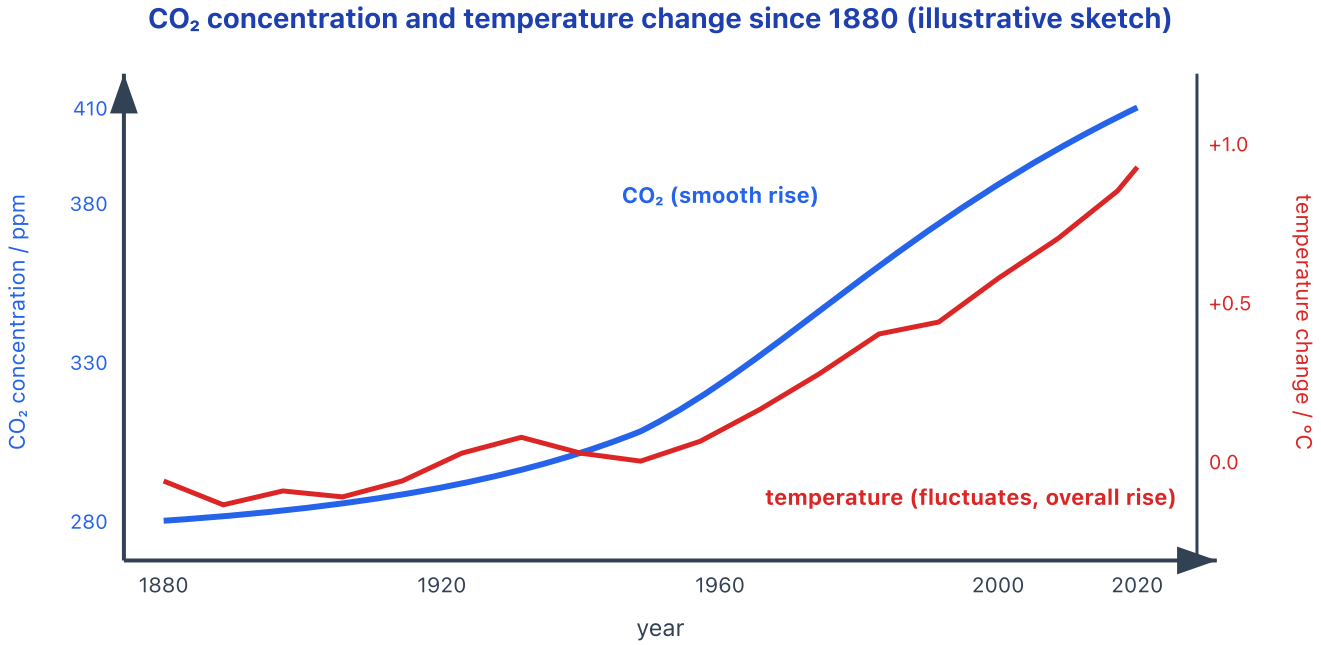
An increase in greenhouse gases enhances the greenhouse effect, causing **global warming** — an increase in the Earth's average surface temperature — which leads to **climate change**.

Evaluating the evidence

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You must be able to **evaluate the evidence for human activity causing climate change**, considering:

- the **correlation** between the change in atmospheric carbon dioxide concentration, the consumption of fossil fuels and temperature change — since the Industrial Revolution all three have risen together;
- the **uncertainties** caused by the **location** where measurements are taken (e.g. readings near cities or volcanoes are not representative; there were few measuring stations in the past, especially over oceans) and the **historical accuracy** of measurements (old thermometers were less accurate; pre-1950s CO_2 values come from indirect methods such as air bubbles in **ice cores**, and older temperatures from tree rings and other proxies).



Illustrative sketch only — shapes show the general trend, values are approximate.

Exam Tip: A **correlation** (two things rising together) does not on its own **prove** that one causes the other. The case for human activity is strong because there is also a **mechanism** (CO_2 absorbs infrared), the vast majority of climate scientists agree after peer review, and computer climate models only match the observed warming when human emissions are included. In an evaluation, mention both the correlation and the uncertainties.

Potential effects of increased CO_2 and methane

- **Rising sea levels** — melting ice caps/glaciers and thermal expansion of seawater → flooding of low-lying land and coastal erosion.
- **Changes in rainfall patterns** and more frequent and severe **extreme weather** (storms, floods, droughts, heatwaves).
- **Changes to the distribution of species** (wildlife habitats shift; some species may become extinct) and to where food crops can be grown — affecting food security.
- **Ocean acidification** — more CO_2 dissolves in seawater, lowering its pH, harming coral reefs and shellfish.

Mitigating the effects

You need to describe how these effects may be mitigated, considering **scale, risk and environmental implications**:

Mitigation	Considerations (scale, risk, environment)
Use **renewable energy** (wind, solar, hydroelectric, tidal) and nuclear instead of fossil fuels	large scale needed; intermittent supply (wind/solar); land use and visual impact; nuclear waste risk
Carbon capture and storage (CCS) — capture CO_2 from power stations and store it underground (e.g. old oil/gas fields)	expensive; technology still developing at scale; risk of leaks
Reforestation / reducing deforestation	trees absorb CO_2 by photosynthesis; needs large land areas; takes decades
Energy efficiency — insulation, efficient appliances, public transport, electric vehicles	relies on individual behaviour; EVs are only low-carbon if electricity is low-carbon
Reduce **methane** — capture landfill gas, reduce food waste, change livestock diets	methane is a more powerful greenhouse gas per molecule, so cuts have a big effect
Use **biofuels** or hydrogen	biofuels compete with food crops for land; hydrogen production may use fossil fuels

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Government action — international agreements, carbon taxes, emissions limits	difficult to agree and enforce globally; economic costs; some countries are still developing

Exam Tip: Six-mark climate questions reward a **balanced, linked** answer: name the effect, explain the chain (e.g. more CO_2 → enhanced greenhouse effect → higher temperature → ice melts → sea level rises), then give a mitigation and a limitation of it.

Key Points Summary

- Hydrocarbons contain carbon and hydrogen only. Crude oil is a complex mixture of hydrocarbons (chains and rings); it is a finite source of fuels and petrochemical feedstock.
- A homologous series has the same general formula, neighbours differ by CH_2, physical properties (e.g. boiling point) change gradually, and chemical properties are similar.
- Fractional distillation: crude oil is vaporised; vapours rise up a column that is cooler at the top; fractions condense at different heights according to boiling point.
- Fractions, top to bottom: gases (heating/cooking), petrol (cars), kerosene (aircraft), diesel oil (some cars, trains), fuel oil (ships, some power stations), bitumen (roads, roofs).
- Larger molecules → higher boiling point, harder to ignite, more viscous (stronger intermolecular forces).
- Complete combustion → $CO_2 + H_2O$ + energy. Limited oxygen → incomplete combustion → CO and/or C (soot).
- CO is toxic: it binds to haemoglobin and reduces the oxygen-carrying capacity of the blood. Faulty appliances produce CO and soot.
- Sulfur impurities → SO_2 → acid rain (damages buildings, lakes, trees). Engine temperatures make N_2 and O_2 from air form NO_x.
- Hydrogen fuel: only water produced, more energy per kg; but storage, safety, infrastructure and how the hydrogen is made are problems.
- Petrol, kerosene and diesel (from crude oil) and methane (natural gas) are non-renewable fossil fuels.

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- Cracking breaks large alkanes into smaller, more useful alkanes and alkenes, using heat and a catalyst; needed because supply of large fractions exceeds demand and alkenes are needed for polymers.
- Early atmosphere (from volcanoes): little or no O_2, lots of CO_2, water vapour, small amounts of other gases. Water vapour condensed to form oceans; CO_2 dissolved in oceans; photosynthesis by primitive plants used CO_2 and released O_2.
- Oxygen relights a glowing splint.
- Greenhouse gases (CO_2, CH_4, H_2O) absorb and re-emit infrared radiation from the Earth, keeping it warm.
- Evaluate evidence for human-caused climate change: correlation of CO_2, fossil fuel use and temperature; uncertainties from measurement location and historical accuracy.
- Effects: rising sea levels, extreme weather, changes to habitats and food production. Mitigation: renewables, CCS, efficiency, reforestation — consider scale, risk and environmental implications.

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